

Charge and Electric Potential

Electric charge

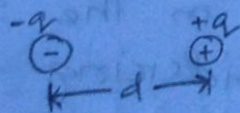
Charge is a fundamental and characteristic property of the elementary particles which make up matter. These particles are Protons, Electrons and Neutrons. Of these, protons are positively charged, electrons negatively and neutrons are neutral. The magnitude of the charge of the proton is the same as those of an electron. There are two kinds of charge, positive and negative.

The smallest amount of charge that has been discovered is that of an electron or of a proton. The magnitude of this charge is 1.602×10^{-19} coulombs.

Electric Field

Electric field is defined as the electric force per unit charge. The direction of the field is taken to be the direction of the force it would exert on a positive test charge. The electric field is radially outward from a positive charge and radially inward toward a negative point charge.

Electric Dipole



When two equal and opposite charges are placed at a small distance apart, then the arrangement is called

an electric dipole.

Electric Dipole Moment

The electric dipole moment for a pair of opposite charges of magnitude q is defined as the magnitude of the charge times the distance between them and the defined direction is toward the positive charge.

Electric dipole moment, $\vec{p} = q\vec{d}$ where d is the distance between the charges.

Electric Potential

The electric potential at a point in an electric field is defined as the work done in bringing a unit positive charge from infinity to that point.

If W be the work done, then the electric potential,

$$V = \frac{W}{q_0}$$

Electric Potential Energy

If the distance between two charges ~~are~~ is increased, an external agent must do work that will be positive if the charges are positive in sign and negative otherwise. The energy represented by this work can be thought stored in the system as electric potential energy. Considering two charges q_1 and q_2 at a distance r apart, the electric potential energy of the system is,

$$U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

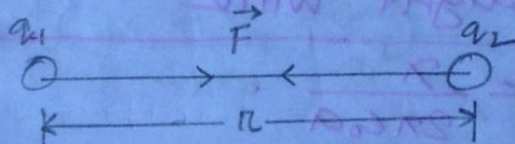
Statement and explanation of Coulomb's law for electrostatics

Statement

The force of attraction or repulsion between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. The force acts along the straight line joining them.

Explanation

Let us consider two point charges q_1 and q_2 at rest at a distance r from each other as shown in figure below.



According to the statement, the mutual force between the two point charges,

$$F \propto q_1 q_2, \text{ when } r \text{ is constant}$$

$$F \propto \frac{1}{r^2}, \text{ when } q_1 \text{ and } q_2 \text{ are constants.}$$

Now, by the rule of proportionality, we can write,

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\Rightarrow F = c \frac{q_1 q_2}{r^2} \quad \text{--- (1)}$$

where C is the constant of proportionality, which is usually written in a more complex way as

$\frac{1}{4\pi\epsilon_0}$, where ϵ_0 is the permittivity constant.

Then the Coulomb's law takes the form,

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \quad (2)$$

In vector notation, equation (2) can be written as,

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^3} \cdot \vec{r}$$

III Derive an expression for electric field at a distance from the uniformly distributed charged long straight wire.

OR, Prove that, $E_p = \frac{\lambda}{2\pi\epsilon_0 a}$

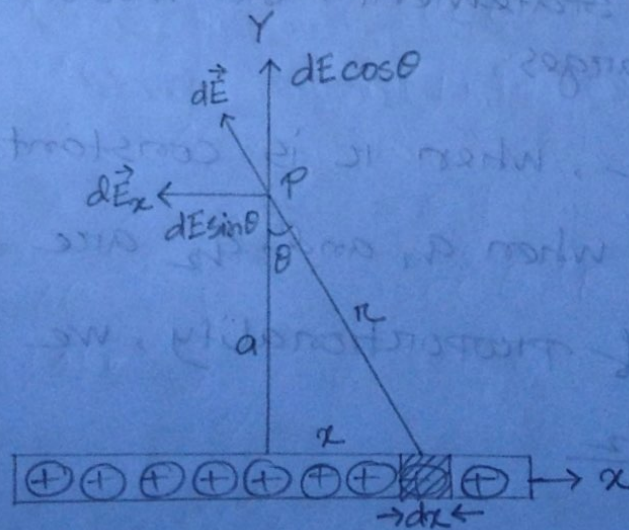


Fig: A section of an infinite line of charge

Let us consider a long uniformly charged wire having charge λ coulombs per meter. So, the magnitude of the electric field at P due to charge $dq = \lambda dx$ is

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$\Rightarrow dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{r^2} \quad \text{--- (1)}$$

The x component of the resultant vector $\vec{E}$ at P must be zero because every charge element on the right has a corresponding element on the left such that their field contributions in the x-direction cancel. Thus only the y component can contribute to the resultant field at P.

$$dE \cos \theta = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{r^2} \cos \theta$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{r^2} \cdot \frac{a}{r}$$

$$= \frac{\lambda a \times dx}{4\pi\epsilon_0 \times r^3} \quad \text{--- (2)}$$

Therefore, the electric field at P is,

$$E_P = \int_{-\infty}^{\infty} dE \cos \theta = \int_{-\infty}^{\infty} \frac{\lambda a}{4\pi\epsilon_0} \cdot \frac{dx}{r^3}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\alpha}^{\alpha} \frac{dx}{(a^2 + x^2)^{3/2}} \quad [\because r = \sqrt{a^2 + x^2}] \quad (3)$$

* Let, $x = a \tan \theta$

$$\Rightarrow dx = a \sec^2 \theta d\theta$$

When, $x = \alpha$, then $\theta = \frac{\pi}{2}$

$x = -\alpha$, then $\theta = -\frac{\pi}{2}$

Hence, eqⁿ (3) becomes

$$E_P = \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2 \theta d\theta}{(a^2 + a^2 \tan^2 \theta)^{3/2}}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2 \theta d\theta}{\{a^2 (1 + \tan^2 \theta)\}^{3/2}}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2 \theta d\theta}{a^3 \sec^3 \theta}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{d\theta}{a^2 \sec \theta}$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} \int_{-\pi/2}^{\pi/2} \cos\theta \, d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} \left[\sin\theta \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} \left[\sin\frac{\pi}{2} - \sin\left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} [1 - (-1)]$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} \times 2$$

$$\therefore E_p = \frac{\lambda}{2\pi\epsilon_0 a} \quad \text{--- (4)}$$

This is the expression for required electric field.

[Note: From eqⁿ (4), it is evident that in the case of a long uniformly charged wire, the electric field is inversely proportional to the first power of the distance from the wire, i.e.

$$E \propto \frac{1}{a}, \text{ where } \frac{\lambda}{2\pi\epsilon_0} = \text{constant} \quad]$$

Q. Derive an expression for the electric field at a point on the axis of a charged circular turn of wire

or, Prove that, $E_P = \frac{q}{4\pi\epsilon_0 b^2}$

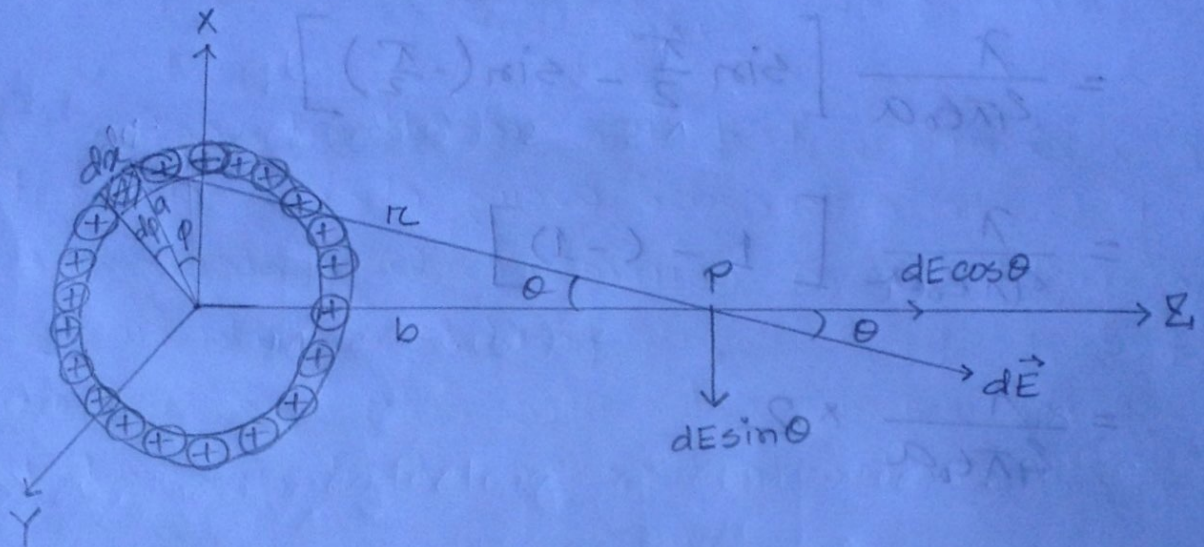


Fig: Electric field at a point on the axis of a circular turn of charged wire.

Let us consider a circular turn of wire of radius a charged with λ coulombs/meter. The electric field at P due to the element of charge

$$dq = \lambda dx = \lambda a d\phi \quad [s = r\phi]$$

is

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$= \frac{\lambda a d\phi}{4\pi\epsilon_0 r^2} \quad \text{--- (1)}$$

From symmetry, the resultant field must lie along the ring axis. Thus only the component of $d\vec{E}$ parallel to this axis contributes to the final result. Therefore, the component,

$$dE \cos \theta = \frac{\lambda a d\phi}{4\pi\epsilon_0 r^2} \cos \theta$$

$$= \frac{\lambda a d\phi}{4\pi\epsilon_0 r^2} \cdot \frac{b}{r}$$

$$\Rightarrow dE \cos \theta = \frac{\lambda ab}{4\pi\epsilon_0 r^3} d\phi \quad \text{--- (2)}$$

Hence, the electric field at P is,

$$E_P = \int_0^{2\pi} dE \cos \theta$$

$$= \int_0^{2\pi} \frac{\lambda ab}{4\pi\epsilon_0 r^3} d\phi$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 r^3} \int_0^{2\pi} d\phi$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 r^3} [\phi]_0^{2\pi}$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 r^3} \times 2\pi$$

$$= \frac{\lambda ab \times 2\pi}{4\pi\epsilon_0 (a^2 + b^2)^{3/2}}$$

$$[\because r = \sqrt{a^2 + b^2}]$$

$$\Rightarrow E_P = \frac{\lambda a b}{2\epsilon_0 (a^2 + b^2)^{3/2}} \quad (3)$$

If the total charge on the ring is $q = 2\pi a \lambda$,

then $\lambda a = \frac{q}{2\pi}$

$$\therefore E_P = \frac{q b}{2\epsilon_0 (a^2 + b^2)^{3/2}} \times \frac{1}{2\pi}$$

$$= \frac{q b}{4\pi\epsilon_0 (a^2 + b^2)^{3/2}} \quad (4)$$

If $b \gg a$, then neglecting a^2 compared ~~to~~ ^{to} b^2 ,

$$E_P = \frac{q b}{4\pi\epsilon_0 b^3} = \frac{q}{4\pi\epsilon_0 b^2}$$

This is the desired expression.

Q. Derive an expression for electric potential at a distance r from a point charge q .

Ans. Prove that $V = \frac{q}{4\pi\epsilon_0 r}$

Let us consider a test charge q_0 is moved from point A to B along the path AB. The field at the point C at a distance $\vec{r}$ from the origin O due to a source charge q at the origin is,

$$\vec{E} = \frac{q}{4\pi\epsilon_0} \cdot \frac{\vec{r}}{r^3} \quad \text{--- (1)}$$

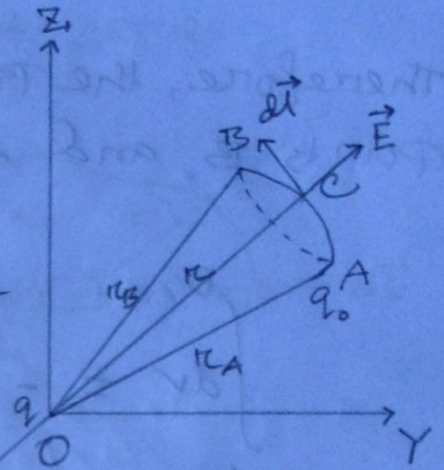


Fig: The path along which the test charge is taken

Let the elementary distance through which the test charge q_0 at C has been displaced be

$d\vec{l}$. Then the elementary work done against the conservative electrostatic force is,

$$\begin{aligned} dW &= -\vec{F} \cdot d\vec{l} \\ &= -q_0 \vec{E} \cdot d\vec{l} \\ &= -\frac{qq_0}{4\pi\epsilon_0} \cdot \frac{\vec{r}}{r^3} \cdot d\vec{l} \quad [\text{From eqn (1)}] \end{aligned} \quad \text{--- (2)}$$

But $\vec{r} \cdot d\vec{l} = r \cdot dr$. Hence from eqn (2),

$$dW = -\frac{qq_0}{4\pi\epsilon_0} \cdot \frac{r dr}{r^3} = -\frac{qq_0}{4\pi\epsilon_0} \cdot \frac{1}{r^2} dr \quad \text{--- (3)}$$

Now, the elementary potential difference is,

$$dv = \frac{dw}{q_0} = \frac{-q q_0}{4\pi\epsilon_0 q_0} \cdot \frac{1}{r^2} dr$$
$$= \frac{-q}{4\pi\epsilon_0 r^2} dr \quad \text{--- (4)}$$

Therefore, the potential difference between the points B and A is,

$$\int_A^B dv = - \int_A^B \frac{q}{4\pi\epsilon_0} \cdot \frac{dr}{r^2}$$

$$\Rightarrow \cancel{V_B - V_A} = \frac{-q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_A^B$$

$$\Rightarrow V_B - V_A = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \quad \text{--- (5)}$$

Let the point A be at infinity, then $V_A = 0$ and $\frac{1}{r_A} = 0$. Denoting r_B by r and V_B by V , eqn (5) yields,

$$V = \frac{q}{4\pi\epsilon_0 r}$$

which is the required expression.

Q. State Gauss's law. Explain its application.

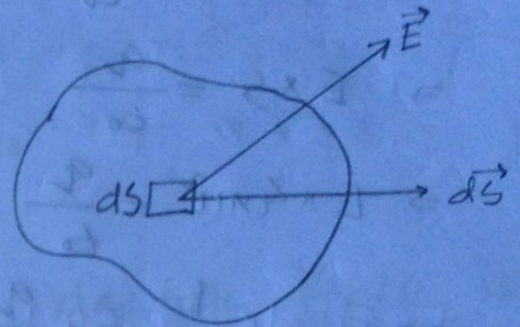
or, Derive Coulomb's law from Gauss's law.

Law Gauss's law states that the flux of electric field $\vec{E}$ through any closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the surface. If the surface is not enclosed, the flux is zero.

According to Gauss's law,

$$\phi = \oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}, \text{ if the surface is closed.}$$

$$\phi = \oint \vec{E} \cdot d\vec{s} = 0, \text{ if the surface is not closed.}$$

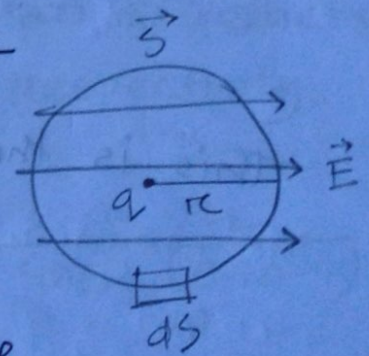


Application of Gauss's Law / Deriving Coulomb's Law

Let us consider a sphere of radius r and q amount of charge is enclosed by the surface.

Area of sphere $S = 4\pi r^2$

Again, let small area $d\vec{s}$ and flux of electric field $\vec{E}$ passing parallelly with $d\vec{s}$ through the enclosed surface.



$$\begin{aligned} \vec{E} \cdot d\vec{s} &= E ds \cos \theta \\ &= E ds \end{aligned}$$

From Gauss's law,

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$\Rightarrow \oint E ds = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \oint ds = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \times S = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \quad \text{--- (1)}$$

We know that, $E = \frac{F}{q_0} \Rightarrow F = q_0 E$

$$\therefore F = \frac{1}{4\pi\epsilon_0} \cdot \frac{qq_0}{r^2}$$

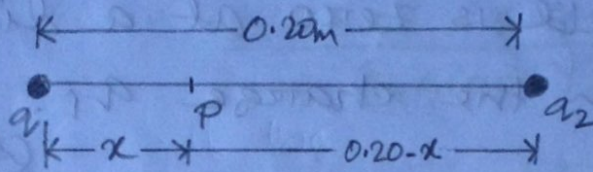
This is the expression for Coulomb's Law.

Mathematical Problems

A positive charge of $q_1 = 2.0 \times 10^{-7} \text{ C}$ is placed at a distance of 0.20 m from another positive charge of $q_2 = 8.0 \times 10^{-7} \text{ C}$. At what point on the line joining them is the electric field zero?

Soln:

[Ans-13]
[SP12-14]



Let P be the point at which the electric field is zero.

x = distance of P from q_1

$0.20 - x$ = distance of P from q_2

Let, E_1 and E_2 be the magnitudes of electric fields at P due to q_1 and q_2 respectively.

$$\therefore E_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{x^2} \quad \text{and} \quad E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_2}{(0.20 - x)^2}$$

Since, electric field at P is zero, hence

$$E_1 = E_2$$

$$\Rightarrow \frac{q_1}{4\pi\epsilon_0 x^2} = \frac{q_2}{4\pi\epsilon_0 (0.20 - x)^2}$$

$$\Rightarrow (0.20 - x)^2 = \frac{q_2}{q_1} x^2$$

$$\Rightarrow (0.20 - x)^2 = \frac{8 \times 10^{-7}}{2 \times 10^{-7}} x^2$$

$$\Rightarrow (0.20 - x)^2 = (2x)^2$$

$$\Rightarrow 3x = 0.20$$

$$\Rightarrow x = \frac{0.20}{3} = 0.067 \text{ m}$$

So, the electric field is zero at a distance of 0.067 m from the charge q_1 .

(Ans)

Q. A dipole is consisting of an electron and proton, $4 \times 10^{-10} \text{ m}$ apart. Compute the electric field at a distance of $2 \times 10^{-8} \text{ m}$ on a line making angle of 45° with the dipole axis from the center of the dipole.

[SPK-14]

Soln:

The magnitude of charge of electron and proton, $q = 1.6 \times 10^{-19} \text{ C}$

distance between electron and proton,

$$2d = 4 \times 10^{-10} \text{ m}$$

distance to compute electric field,

$$r = 2 \times 10^{-8} \text{ m}$$

$$\text{angle, } \theta = 45^\circ$$

Dipole moment, $P = q \times 2d$

$$= 1.6 \times 10^{-19} \times 4 \times 10^{-10}$$

$$= 6.4 \times 10^{-29} \text{ Cm}$$

$$\therefore \text{Electric field, } E = \frac{P}{4\pi\epsilon_0 r^3} \sqrt{1+3\cos^2\theta}$$

$$= \frac{9 \times 10^9 \times 6.4 \times 10^{-29}}{(2 \times 10^{-8})^3} \sqrt{1+3\cos^2 45^\circ}$$

$$= 1.138 \times 10^5 \text{ NC}^{-1}$$

(Am)

Q. A thin circular ring of radius a meter carries a total positive charge of q coulomb. Calculate the electric field at the center of ring.

Soln:

Given,

amount of charge = q C

radius = a m

~~distance~~

We know that,

$$E = \frac{qb}{4\pi\epsilon_0 (a^2 + b^2)^{3/2}}$$

Here, distance from the center, $b = 0$

$$\text{So, } E = \frac{q \times 0}{4\pi\epsilon_0 (a^2 + 0^2)^{3/2}} = 0 \text{ NC}^{-1}$$

(Am)

Q A cylinder of large length has a charge of $2 \times 10^{-8} \text{ Cm}^{-1}$. Find the electric field at a distance of 0.2 m from it.

[Aut-14]

[Aut-15]

Solⁿ:

Given,

charge per length, $\lambda = 2 \times 10^{-8} \text{ Cm}^{-1}$

distance, $a = 0.2 \text{ m}$

$$\therefore \text{Electric field, } E = \frac{\lambda}{2\pi\epsilon_0 a}$$

$$= \frac{2\lambda}{4\pi\epsilon_0 a}$$

$$= \frac{(9 \times 10^9) \times 2 \times (2 \times 10^{-8})}{0.2}$$

$$= 1800 \text{ N C}^{-1}$$

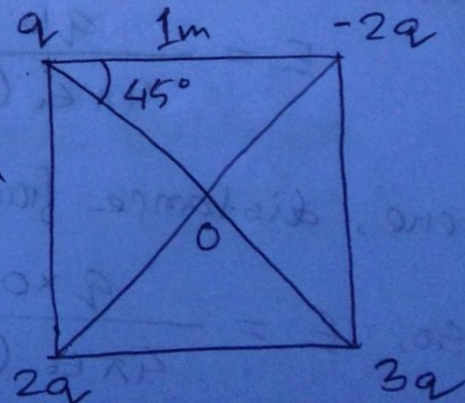
(Am)

Q Find the potential at the center of a 1.0 m square having charges $q, -2q, 3q$ and $2q$ at its corners. ($q = 1.0 \times 10^{-8} \text{ C}$). [Aut-14] [Aut-13]

Solⁿ:

The center point O is at a distance of $r = \frac{1\sqrt{2}}{2} = 0.71 \text{ m}$ from each corner.

The potential at O due to all the four charges is,



$$V = \frac{1 \times q}{4\pi\epsilon_0 \times r} + \frac{1 \times -2q}{4\pi\epsilon_0 \times r} + \frac{1 \times 3q}{4\pi\epsilon_0 \times r} + \frac{1 \times 2q}{4\pi\epsilon_0 \times r}$$

$$= \frac{1}{4\pi\epsilon_0 r} [q - 2q + 3q + 2q]$$

$$= \frac{4q}{4\pi\epsilon_0 r}$$

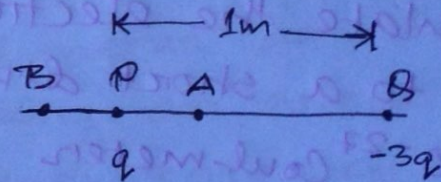
$$= (9 \times 10^9) \times \frac{4 \times 1 \times 10^{-8}}{0.71}$$

$$= 507.04 \text{ V} \quad (\text{Ans})$$

Q Two charges $+q$ and $-3q$ are separated by a distance of 1m . At what points on its axis is the potential zero? [SPIC-14]

Solⁿ:

Let the potential be zero at A and B.



(i) The point A is on the right side of $+q$ at a distance of x . So, its distance from $-3q$ is $(1-x)$.

For $V=0$, we require $\frac{q}{4\pi\epsilon_0 x} + \frac{-3q}{4\pi\epsilon_0 (1-x)} = 0$

$$\text{or, } \frac{q}{4\pi\epsilon_0} \left(\frac{1}{x} - \frac{3}{1-x} \right) = 0$$

$$\text{or, } \frac{1}{x} = \frac{3}{1-x}$$

$$\text{or, } 3x = 1-x$$

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$\therefore x = 0.25 \text{ m}$, on the right of $+q$.

(ii) The point B is on the left of $+q$ at a distance y . Its distance from $-3q$ is $(1+y)$.

The potential at B due to $+q$ and $-3q$ is 0.

So,

$$\frac{q}{4\pi\epsilon_0 y} + \frac{-3q}{4\pi\epsilon_0 (1+y)} = 0$$

$$\Rightarrow \frac{1}{y} = \frac{3}{1+y}$$

$$\Rightarrow 3y = 1+y$$

$\therefore y = 0.5 \text{ m}$, on the left of $+q$.

(Am)

Q calculate the electric potential and field due to a short dipole of dipole moment $2 \times 10^{-27} \text{ Coul-meter}$ at a point distant 2 cm from on its angle 90° .

[SPR-13]

Soln:

Given,

dipole moment, $P = 2 \times 10^{-27} \text{ Cm}$

distance, $r = 2 \text{ cm} = 0.02 \text{ m}$

angle, $\theta = 90^\circ$

So,

$$\text{Electric potential, } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{P \cos\theta}{r^2}$$

$$= 9 \times 10^9 \times \frac{2 \times 10^{-27} \times \cos 90^\circ}{(0.02)^2}$$

$$= 0 \text{ V } [\because \cos 90^\circ = 0]$$

$$\text{and Electric field, } E = \frac{1}{4\pi\epsilon_0} \times \frac{P}{r^3} \sqrt{1 + 3 \cos^2\theta}$$

$$= 9 \times 10^9 \times \frac{2 \times 10^{-27}}{(0.02)^3} \times \sqrt{1 + 3 \times 0}$$

$$= 2.25 \times 10^{-12} \text{ N C}^{-1} \text{ (Am)}$$

Q Calculate the Potential due to a short dipole of dipole moment 2×10^{-27} Coul-meter at a distance of 2×10^{-8} m on a line making an angle 45° with the dipole axis from the center of the dipole.

[Ans-13]

Soln: Given,

$$\text{dipole moment, } P = 2 \times 10^{-27} \text{ Cm}$$

$$\text{distance, } r = 2 \times 10^{-8} \text{ m}$$

$$\text{angle, } \theta = 45^\circ$$

$$\therefore \text{Potential, } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cos\theta}{r^2}$$

$$= 9 \times 10^9 \times \frac{2 \times 10^{-27} \times \cos 45^\circ}{(2 \times 10^{-8})^2}$$

$$= 31.82 \times 10^{-3} \text{ V}$$

(Am)

Q Two protons in a nucleus of U^{238} are 6.0×10^{-15} meter apart. What is their mutual electric potential energy? [Ans - 15]

Solⁿ:

charges, $q_1 = q_2 = 1.6 \times 10^{-19} \text{ C}$

distance, $r = 6 \times 10^{-15} \text{ m}$

$\therefore \text{Energy} = \text{Work done, } W = q_2 V$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

$$= 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{6 \times 10^{-15}}$$

$$= 3.84 \times 10^{-14} \text{ J}$$

(Am)

Q The atomic number of gold is 79 and the charge on proton is $1.6 \times 10^{-19} \text{ C}$. Calculate the electrical potential at the surface of the nucleus of the gold atom. The radius of the nucleus is $6.6 \times 10^{-15} \text{ m}$.

Solⁿ:

Given,

$$\begin{aligned}\text{charge, } q &= 79 \times 1.6 \times 10^{-19} \text{ C} \\ &= 1.264 \times 10^{-17} \text{ C}\end{aligned}$$

$$\text{radius, } r = 6.6 \times 10^{-15} \text{ m}$$

$$\therefore \text{Electrical Potential, } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

$$= 9 \times 10^9 \times \frac{1.264 \times 10^{-17}}{6.6 \times 10^{-15}}$$

$$= 1.7 \times 10^7 \text{ V (Ans)}$$

MAGNETIC FIELD

Magnetic field

The space around the current carrying conductor is defined as the site of magnetic field. Magnetic field is a vector field whose magnitude and direction at any point specified by a vector $\vec{B}$ called magnetic induction.

Magnetic flux

The magnetic flux Φ through a surface S is defined as $\Phi = \int \vec{B} \cdot d\vec{s}$

It represents the lines of induction crossing surface S .

The SI unit of magnetic flux is Wb (Weber).

Magnetic flux density

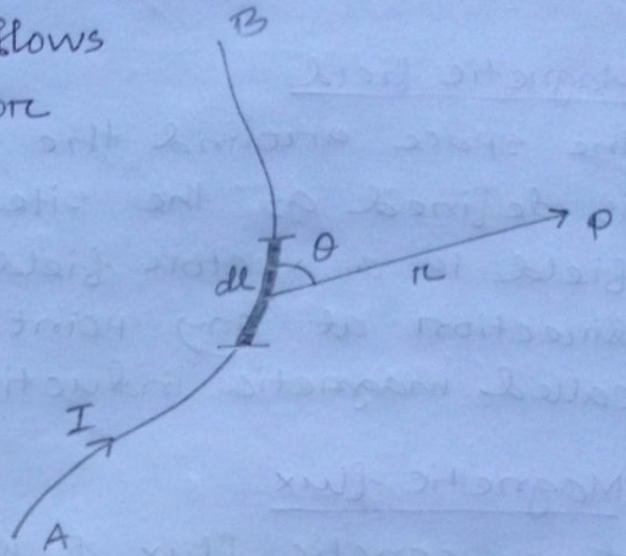
The amount of flux passing through a unit area at right angles to the magnetic field is called the magnetic flux density.

So, the magnetic flux density, $B = \frac{\Phi}{A}$

The SI unit of magnetic flux density is Wb m^{-2} or T (Tesla).

State and explain Biot-Savart's law.

Statement: When a current flows through a small element for a particular media, the magnetic field is directly proportional to the length, the strength of current, sine of an angle between the conductor and the line joining the point and inversely proportional to the square of the distance of the given point from the conductor.



Explanation:

Let us consider current I is passing through the conductor AB and dl is a small portion of AB . If the distance between dl and any point P be r and angle is θ , then the magnetic field at P ,

$$dB \propto \frac{I dl \sin \theta}{r^2}$$

$$\Rightarrow dB = K \frac{I dl \sin \theta}{r^2} \quad \text{--- (1)}$$

Here, K is the constant of proportionality. In vacuum medium, $K = \frac{\mu_0}{4\pi}$ where μ_0 is permeability of vacuum.

So, for vacuum medium, we can write,

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin \theta}{r^2}$$

Magnetic induction/field at a point due to a straight conductor carrying current

Let us consider a straight conductor XY carrying a current I in the direction X to Y. We will calculate magnetic induction at the point P.

Let,

$SP = a$ = Perpendicular distance between conductor and P

dl = length of a small element of the conductor distant l from S

r = distance between dl and P

I = flow of current through XY

θ = angle between current flow and OP

According to Biot-Savart's law, magnetic induction at P due to dl ,

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin \theta}{r^2} \quad \text{--- (1)}$$

$\therefore$ Magnetic field at P due to the whole conductor,

$$B = \int dB = \int_{l=-\infty}^{l=+\infty} \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin \theta}{r^2}$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi} \int_{l=-\infty}^{l=+\infty} \frac{dl \sin \theta}{r^2} \quad \text{--- (2)}$$

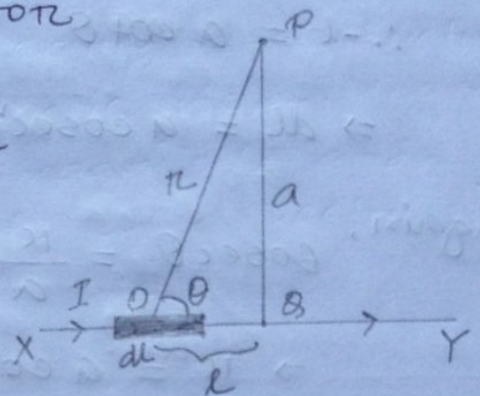


Fig: Conductor carrying current

Here,

$$\cot \theta = \frac{-l}{a}$$

$$\therefore -l = a \cot \theta$$

$$\Rightarrow dl = a \operatorname{cosec}^2 \theta d\theta$$

again, $\operatorname{cosec} \theta = \frac{\pi}{a}$

$$\Rightarrow \pi = a \operatorname{cosec} \theta$$

When $l = -\infty$, then $\theta = 0$

$l = +\infty$, then $\theta = \pi$

So, from exp. (2),

$$B = \frac{\mu_0 I}{4\pi} \int_0^\pi \frac{a \operatorname{cosec}^2 \theta \sin \theta}{a^2 \operatorname{cosec}^2 \theta} d\theta$$

$$= \frac{\mu_0 I}{4\pi a} \int_0^\pi \sin \theta d\theta$$

$$= -\frac{\mu_0 I}{4\pi a} [\cos \theta]_0^\pi$$

$$= -\frac{\mu_0 I}{4\pi a} [\cos \pi - \cos 0]$$

$$= -\frac{\mu_0 I}{4\pi a} \times -2$$

$$\therefore B = \frac{\mu_0 I}{2\pi a}$$

Magnetic induction at a point on the axis of a circular coil carrying current

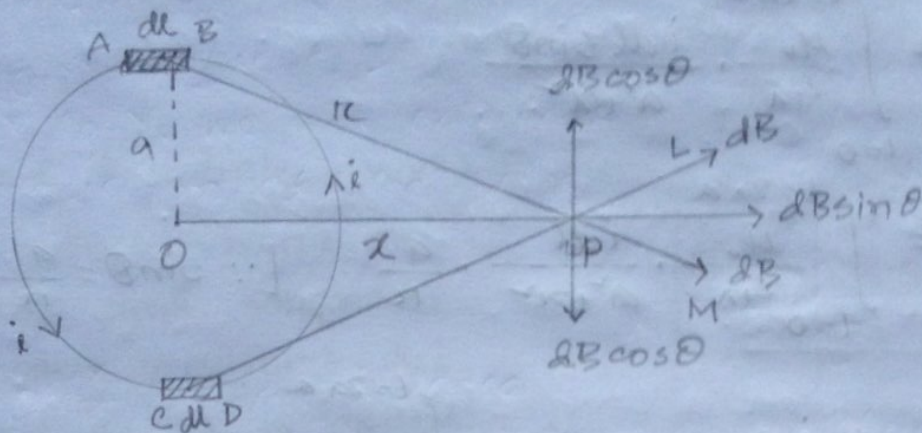


Fig: Circular coil carrying current

Let us consider a circular coil of radius a , carrying current i . P is a point on its axis at a distance x from the center O . Let AB and CD be two opposite current elements each of same length dl . The distance of P from any point on the circumference of the coil is $r = \sqrt{a^2 + x^2}$.

According to Biot-Savart's law, the field at P due to AB is,

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{idl \sin \theta}{r^2}$$

Here, $\theta = 90^\circ$ and $r^2 = a^2 + x^2$. Putting these,

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{idl}{a^2 + x^2} \quad \text{--- (1)}$$

the magnetic field at P due to CD is also $\frac{\mu_0}{4\pi} \cdot \frac{idl}{a^2 + x^2}$

The components $dB \cos \theta$ cancel each other. Thus, the total magnetic induction at P due to the entire coil,

$$B = \int_{\lambda=0}^{\lambda=2\pi a} dB \sin \theta$$

$$= \int_{\lambda=0}^{\lambda=2\pi a} \frac{\mu_0}{4\pi} \cdot \frac{idl \sin \theta}{a^2 + x^2}$$

$$= \int_{\lambda=0}^{\lambda=2\pi a} \frac{\mu_0}{4\pi} \cdot \frac{idl}{a^2 + x^2} \cdot \frac{a}{a} \quad [\because \sin \theta = \frac{a}{r}]$$

$$= \frac{\mu_0 ia}{4\pi (a^2 + x^2) \sqrt{a^2 + x^2}} \int_{\lambda=0}^{\lambda=2\pi a} dl \quad [\because r = \sqrt{a^2 + x^2}]$$

$$= \frac{\mu_0 ia}{4\pi (a^2 + x^2)^{3/2}} \times [l]_0^{2\pi a}$$

$$= \frac{\mu_0 ia}{4\pi (a^2 + x^2)^{3/2}} \times 2\pi a$$

$$\therefore B = \frac{\mu_0 i a^2}{2(a^2 + x^2)^{3/2}}$$

This is the required magnetic field due to a circular coil loop.

Case 1: If the coil has N turns, then

$$B = \frac{\mu_0 i a^2 N}{2(a^2 + x^2)^{3/2}}$$

Case 2: At the center of the coil, $x = 0$, then

$$B = \frac{\mu_0 i N}{2a}$$

Case 3: When $x \gg a$, then

$$B = \frac{\mu_0 i a^2 N}{2x^3}$$

Lorentz Force

If a charged particle moves through a region, where an electric field and magnetic field both are present, the resultant force is called Lorentz force and is denoted by $\vec{F}_L$.

Let us consider a charge q moving with a velocity $\vec{v}$ where magnetic field $\vec{B}$ and electric field $\vec{E}$ are present.

Electric field, $\vec{F}_1 = q\vec{E}$ and Magnetic field, $\vec{F}_2 = q\vec{v} \times \vec{B}$

$$\begin{aligned}\therefore \text{Lorentz Force, } \vec{F}_L &= \vec{F}_1 + \vec{F}_2 \\ &= q\vec{E} + q\vec{v} \times \vec{B} \\ &= q(\vec{E} + \vec{v} \times \vec{B})\end{aligned}$$

This equation is called Lorentz force equation.

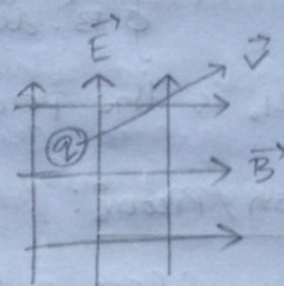


Fig: A moving charge

State and explain Ampere's circuital law.

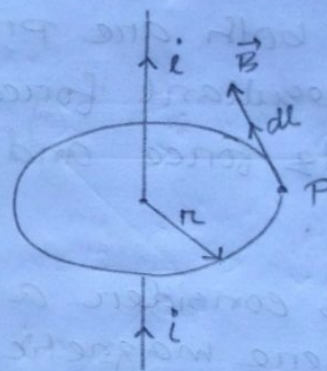
Statement: The line integral $\oint \vec{B} \cdot d\vec{l}$ for a closed curve is equal to μ_0 times the net current i through the area bounded by the curve i.e.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

where μ_0 is the permeability constant.

Explanation/Proof:

Let us consider a long straight conductor carrying a current i perpendicular to the page directed outward as shown in figure. According to Biot-Savart's law, the net magnitude of the magnetic induction at a distance r from it is given by,



$$B = \frac{\mu_0 i}{2\pi r} \quad \text{--- (1)}$$

At each point of the circle, $\vec{B}$ has a constant magnitude B and $d\vec{l}$ which is always tangential to the path of integration, points in the same direction as $\vec{B}$, thus

$$\oint \vec{B} \cdot d\vec{l} = \oint B \cdot dl = B \oint dl = B (2\pi r)$$

Here $\oint dl = 2\pi r$ is the circumference of the circle.

Substituting the value of B from eqⁿ (1), we get

$$\oint \vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2\pi r} \cdot 2\pi r = \mu_0 i$$

(proved)

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Advantages of Biot-Savart's law over Ampere's law

Ampere's law is usually applied to problems those have certain symmetry. For example, a straight wire is symmetric about its center axis, a sheet is symmetric, a solenoid is symmetric etc. But some arbitrary wire is not symmetric however and Ampere's law can't be applied there generally.

If we consider a short solenoid or wire (a short length in relation to its radius), Ampere's law will not give accurate answer either, because it ignores the fact that the magnetic field lines curve away from the wire at the ends, it assumes the magnetic field is nearly constant.

MATHEMATICAL PROBLEMS

Find the magnetic induction at the center of a square current loop of side d unit carrying a current i unit.

Or, Find the magnetic induction at the center of a square current loop of side 1m carrying a current 1 ampere.

[SPN-2013]

Soln;

Let ABCD be the square current loop of side d meter carrying current i ampere and P be the center point at which the magnetic induction due to ABCD is to be calculated.

Let O be the mid point of AB .

So, $OP = \frac{d}{2}$ meter and

$\angle BPO = \angle APO = 45^\circ$.

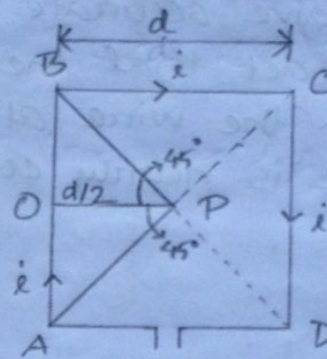
Now, magnetic field at P due to AB ,

$$B' = \frac{\mu_0}{4\pi} \cdot \frac{i}{a} (\sin \angle BPO + \sin \angle APO)$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{i}{\frac{d}{2}} (\sin 45^\circ + \sin 45^\circ)$$

$$= \frac{\mu_0 \times 2i}{4\pi \times d} \times \frac{2}{\sqrt{2}}$$

$$= \frac{\mu_0 i}{\sqrt{2} \pi d}$$



the magnetic induction due to current in the remaining three arms of the square loop will also have the same magnitude and direction. Thus, the total magnetic induction at P,

$$B = 4B' = 4 \times \frac{\mu_0 i}{\sqrt{2}\pi d} = \frac{2\sqrt{2}\mu_0 i}{\pi d} \quad (\text{Am})$$

OR

Given, $d = 1\text{m}$

$i = 1\text{A}$

We know, $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$

$$\text{So, } B = \frac{2\sqrt{2} \times 4\pi \times 10^{-7} \times 1}{\pi \times 1}$$

$$= 8\sqrt{2} \times 10^{-7} \text{ T (or Wbm}^{-2}\text{)} \quad (\text{Am})$$

Q A circular coil has a radius of 0.25 m and number of turns 100. Calculate the magnetic field at a point on the axis of the coil and distance 1m from the center, when a current of 0.2A flows in it.

[Ans-15]

Soln:

Given,

radius, $a = 0.25\text{m}$

turns, $N = 100$

distance, $x = 1\text{m}$

current, $i = 0.2\text{A}$

We know that,

$$B = \frac{\mu_0 i a^2 N}{2(a^2 + x^2)^{3/2}} = \frac{4\pi \times 10^{-7} \times 0.2 \times (0.25)^2 \times 100}{2\{(0.25)^2 + 1^2\}^{3/2}}$$

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$$= 7.17 \times 10^{-7} \text{ T (or Wbm}^{-2}\text{)}$$

(Am)

[Aut-14]

Given, radius, $a = 0.15 \text{ m}$

turns, $N = 66$

distance, $x = 0.25 \text{ m}$

current, $i = 0.15 \text{ A}$

$$\begin{aligned} \text{Magnetic field, } B &= \frac{\mu_0 i a^2 N}{2(a^2 + x^2)^{3/2}} = \frac{4\pi \times 10^{-7} \times 0.15 \times (0.15)^2 \times 66}{2(0.15^2 + 0.25^2)^{3/2}} \\ &= 5.648 \times 10^{-6} \text{ T (or Wbm}^{-2}\text{)} \end{aligned}$$

(Am)

[Aut-13]

Given,

radius, $a = 0.1 \text{ m}$

turns, $N = 55$

current, $i = 0.15 \text{ A}$

We know that magnetic field at the center,

$$\begin{aligned} B &= \frac{\mu_0 i N}{2a} = \frac{4\pi \times 10^{-7} \times 0.15 \times 55}{2 \times 0.1} \\ &= 5.18 \times 10^{-5} \text{ T (or Wbm}^{-2}\text{)} \end{aligned}$$

(Am)

Magnetic Field

(More mathematical problems)

- Q A circular coil has a radius of 0.1 m and a number of turns of 50. Calculate the magnetic field at a point
(i) on the axis of the coil and distance 0.2 m from the center (ii) at the center of the coil, when a current of 0.1 A flows in it.

Solⁿ:

(i) Here,

radius, $a = 0.1 \text{ m}$

turns, $N = 50$

distance, $x = 0.2 \text{ m}$

current, $i = 0.1 \text{ A}$

$$\therefore B = \frac{\mu_0 N i a^2}{2(a^2 + x^2)^{3/2}} = \frac{4\pi \times 10^{-7} \times 50 \times 0.1 \times (0.1)^2}{2\{(0.1)^2 + (0.2)^2\}^{3/2}}$$
$$= 2.81 \times 10^{-6} \text{ T (or } \text{Wb m}^{-2}) \text{ (Am)}$$

(ii) At the center, $x = 0 \text{ m}$.

$$\text{So, } B = \frac{\mu_0 N i}{2a} = \frac{4\pi \times 10^{-7} \times 50 \times 0.1}{2 \times 0.1} = 3.14 \times 10^{-5} \text{ T (Am)}$$

Q In the Bohr model of hydrogen atom, the electron circulates around the nucleus in a path of radius $5.29 \times 10^{-11} \text{ m}$ at a frequency of $6.58 \times 10^{15} \text{ Hz}$. Find the magnitudes of the magnetic induction at the center of the orbit. What is its dipole moment?

Solⁿ: We know that,

$$\text{Current} = \frac{\text{Charge}}{\text{Time}} = \frac{e}{T} = ef$$

where e is the electronic charge and f is the frequency of revolution $\left[\text{time} = \frac{1}{\text{frequency}} \right]$,

$$\text{i.e. } T = \frac{1}{f} \text{ or } f = \frac{1}{T}$$

$$\therefore I = ef = 1.602 \times 10^{-19} \times 6.58 \times 10^{15} \\ = 1.054 \times 10^{-3} \text{ A}$$

So, magnetic induction at the center of the orbit,

$$B = \frac{\mu_0 I}{2a} = \frac{4\pi \times 10^{-7} \times 1.054 \times 10^{-3}}{2 \times 5.29 \times 10^{-11}} = 12.52 \text{ T} \quad (\text{Am})$$

Area of the circular orbit, $A = \pi a^2$

So, magnetic dipole moment,

$$M = AI = \pi a^2 I = \pi \times (5.29 \times 10^{-11})^2 \times 1.054 \times 10^{-3} \\ = 9.27 \times 10^{-24} \text{ Am}^2 \quad (\text{Am})$$

Q A long straight vertical wire carries a current of 30 A directed upwards. Calculate the position of neutral point if the horizontal component of earth's magnetic field is $2 \times 10^{-5} \text{ Wbm}^{-2}$.

Soln: Given,

$$B_H = 2 \times 10^{-5}$$

$$I = 30 \text{ A}$$

We know that,

$$B_H = \frac{\mu_0 I}{2\pi a}$$

$$\therefore a = \frac{\mu_0 I}{2\pi B_H} = \frac{4\pi \times 10^{-7} \times 30}{2\pi \times 2 \times 10^{-5}} = 0.3 \text{ m (Am)}$$

Q What is the magnetic field intensity at a distance of 5 cm due to a long straight conductor carrying a current of 150 A?

Soln: Given,

$$\text{distance, } a = 5 \text{ cm} = 0.05 \text{ m}$$

$$\text{current, } i = 150 \text{ A}$$

$$\text{So, } B = \frac{\mu_0 i}{2\pi a} = \frac{4\pi \times 10^{-7} \times 150}{2\pi \times 0.05} = 6 \times 10^{-4} \text{ T (Am)}$$

Q The magnetic field at a distance 2.3 cm from the axis of a long straight wire is 13 mT. What is the current in the wire?

Soln:

Given,

$$\text{distance, } a = 2.3 \text{ cm} = 0.023 \text{ m}$$

$$\text{magnetic field, } B = 13 \text{ mT} = 13 \times 10^{-3} \text{ T}$$

So, $B = \frac{\mu_0 i}{2\pi a}$

$$\Rightarrow i = \frac{B \times 2\pi a}{\mu_0} = \frac{13 \times 10^{-3} \times 2\pi \times 0.023}{4\pi \times 10^{-7}}$$

$$= 1495 \text{ A}$$

(Am)

ELECTROMAGNETIC INDUCTION

Q1 State the Faraday's laws of electromagnetic induction.

Law 1: Whenever the magnetic flux through a conductor is changed, an emf is induced in the conductor. The magnitude of the induced emf is equal to the rate of change of magnetic flux through the circuit. If ϕ is the magnetic flux linked with the circuit at any instant t and $\mathcal{E}$ is the induced emf, then

$$\mathcal{E} \propto \frac{d\phi}{dt}$$

Law 2: The direction of the induced emf or current is such as to oppose the change that produced it. This is known as Lenz's law.

Combining both laws, $\mathcal{E} = -\frac{d\phi}{dt}$

Self Induction

The phenomenon of the production of an induced emf in a circuit itself due to the change in current through it is called self induction and the induced emf is called back emf.

Self Inductance

The magnetic flux ϕ produced in a coil is directly proportional to the current I flowing in the coil, i.e.

$$\phi \propto I$$

$$\Rightarrow \phi = LI$$

where L is the constant of proportionality, called the

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co-efficient of self induction or self inductance of the coil.

SI unit of self inductance is H (henry).

Self-inductance of a Long Solenoid

Let us consider a long-coiled solenoid where,

l = length of solenoid

N = total number of turns

N/l = turns per unit length

I = current flow through it

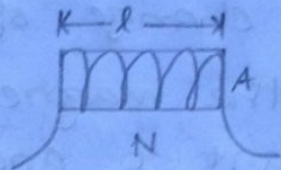


Fig : Solenoid

So, magnetic field inside it, $B = \mu_0 \left(\frac{N}{l} \right) I$

and Flux through each turn = ϕ_B

$$= BA$$

$$= \frac{\mu_0 N I A}{l}$$

Total flux through N turns, $\phi = \phi_B \times N$

$$= \frac{\mu_0 N^2 I A}{l}$$

Again,

self inductance of the solenoid,

$$L = \frac{\phi}{I} = \frac{\mu_0 N^2 I A}{l} / I$$

$$\Rightarrow L = \frac{\mu_0 N^2 A}{l}$$

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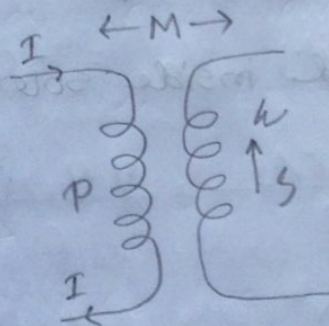
[Note: If the solenoid is wound on a core of constant permeability μ , then

$$L = \frac{\mu N^2 A}{l}, \text{ where } \mu = \mu_0 \mu_r]$$

Mutual Inductance

If the current flowing in a coil changes, the magnetic flux linked with another neighbouring coil also changes, hence an induced emf is set up in the second coil. This phenomenon is called mutual inductance.

The circuit in which the current changes is called primary circuit and the neighbouring circuit in which the emf is induced is called secondary circuit.



Let I be the current in primary coil. The magnetic flux through secondary coil depends upon I .

$$\text{So, } \phi \propto I$$

$$\Rightarrow \phi = MI$$

where M is the co-efficient of mutual induction or simply mutual inductance of the two circuits.

The SI unit of mutual inductance is henry (H).

Mutual Inductance between two Coaxial Solenoids.

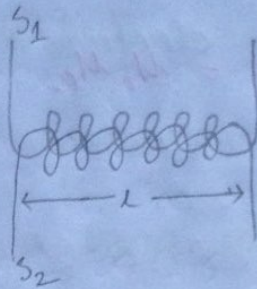


Fig : Two coaxial solenoids

Let us consider two long coaxial solenoids S_1 and S_2 of same length l and area of cross-section A .

N_1 = number of turns in primary S_1

N_2 = number of turns in secondary S_2

I = current passing through S_1

So, magnetic field produced inside solenoid S_1 ,

$$B = \frac{\mu_0 N_1 I}{l}$$

$$\Rightarrow BA = \frac{\mu_0 N_1 I A}{l}$$

$$\text{thus, magnetic flux, } \phi = \frac{\mu_0 N_1 I A}{l} \times N_2 = \frac{\mu_0 N_1 N_2 I A}{l}$$

We know that, $\phi = MI$

$$\therefore M = \frac{\phi}{I} = \frac{\mu_0 N_1 N_2 A}{l}$$

This is the desired expression.

[Note: For other materials, $M = \frac{\mu N_1 N_2 A}{l}$, where

$\mu = \mu_0 \mu_r$ and μ_r = permeability for the material.]

Mathematical Problems

Ex A solenoid having an air core and 10 cm long has 100 turns and its area of cross-section is 5 sq. cm. Find the co-efficient of self induction of the solenoid. [SPM-13]

Solⁿ: Given,

$$\text{length, } l = 10 \text{ cm} = 0.1 \text{ m}$$

$$\text{area, } A = 5 \text{ cm}^2 = 5 \times 10^{-4} \text{ m}^2$$

$$\text{turns, } N = 100$$

$$\text{We know that, } L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 100^2 \times 5 \times 10^{-4}}{0.1}$$

$$= 6.28 \times 10^{-5} \text{ henry}$$

(Am)

Ex Calculate the self inductance of a solenoid having 1000 turns and length 1m. the area of cross-section is 7 cm² and the relative permeability of the core is 1000.

Solⁿ: Given,

$$\text{turns, } N = 1000$$

$$\text{length, } l = 1 \text{ m}$$

$$\text{area, } A = 7 \text{ cm}^2 = 7 \times 10^{-4} \text{ m}^2$$

$$\text{relative permeability, } \mu_r = 1000$$

We know that,

$$L = \frac{\mu_0 \mu_r N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 1000 \times 1000^2 \times 7 \times 10^{-4}}{1}$$

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$$= 0.88 \text{ henry} \quad (\text{Am})$$

Ex A solenoid having a core of cross section 4cm^2 , half air and half iron (relative permeability 500) is 22 cm long. If the number of turns on it is 1000, what will be its self-inductance?

[SPRC-14]

Solⁿ: Here,

area of cross section for air and iron,

$$A_1 = A_2 = \frac{4\text{cm}^2}{2} \quad [\text{because half air, half iron}]$$

$$= 2 \times 10^{-4} \text{ m}^2$$

$$\text{turns, } N = 1000$$

$$\text{length, } l = 22 \text{ cm} = 0.22 \text{ m}$$

$$\text{relative permeability, } \mu_r = 500$$

$$\text{So, self inductance, } L = \frac{\mu_0 N^2 A_1}{l} + \frac{\mu_0 \mu_r N^2 A_2}{l}$$

$$= \frac{\mu_0 N^2}{l} (A_1 + \mu_r A_2)$$

$$= \frac{4\pi \times 10^{-7} \times 1000^2}{0.22} \left\{ (2 \times 10^{-4}) + (500 \times 2 \times 10^{-4}) \right\}$$

$$= 0.5 \text{ henry} \quad (\text{Am})$$

Q A solenoid having a core of cross section 6cm^2 and 25cm long. If the number of turns on it is 500, what will be its self inductance?

[Ans -15]

Soln:

$$L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 500^2 \times 6 \times 10^{-4}}{0.25}$$

$$= 7.54 \times 10^{-4} \text{ henry (Am)}$$

Q A solenoid of length 30cm and area of cross-section 10sq.cm has 1000 turns wound over a core of constant permeability 600. Another coil of 500 turns is wound over the same coil at its middle. Calculate the mutual inductance between them.

Soln:

Given,

turns, $N_1 = 1000$

turns, $N_2 = 500$

permeability, $\mu_r = 600$

length, $l = 30\text{cm} = 0.3\text{m}$

area, $A = 10\text{cm}^2 = 10^{-3}\text{m}^2$

We know that, mutual inductance,

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{l} = \frac{4\pi \times 10^{-7} \times 600 \times 1000 \times 500 \times 10^{-3}}{0.3}$$

$$= 1.26 \text{ henry (Am)}$$

Q A solenoid of length 0.35 m and area of cross section $10 \times 10^{-4} \text{ m}^2$ has 1000 turns wound over a core of constant permeability 600. Another coil of 550 turns is wound over the same coil at its middle. Calculate the mutual inductance between them.

[Aut-13]

Soln: Here,

$$l = 0.35 \text{ m}$$

$$A = 10 \times 10^{-4} \text{ m}^2$$

$$N_1 = 1000$$

$$N_2 = 550$$

$$\mu_r = 600$$

$$\therefore M = \frac{\mu_r \mu_0 N_1 N_2 A}{l} = \frac{4\pi \times 10^{-7} \times 600 \times 1000 \times 550 \times 10 \times 10^{-4}}{0.35}$$

$$= 1.18 \text{ Henry (Am)}$$

Q Two coils, a primary of 900 turns and a secondary of 45 turns, are wound on an iron ring of mean radius 0.1 m and cross section $4 \times 10^{-2} \text{ m}$ diameter. Find their mutual inductance (μ_r for iron = 800).

[Aut-14]

Soln:

Here,

$$\text{length, } l = 2\pi \times 0.1 \text{ m} = 0.2\pi \text{ m}$$

$$\text{cross-section diameter, } d = 4 \times 10^{-2} \text{ m}$$

$$\therefore \text{radius} = \frac{d}{2} = 2 \times 10^{-2} \text{ m}$$

$$\therefore \text{area of cross section, } A = \pi \left(\frac{d}{2}\right)^2$$

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$$= \pi (2 \times 10^{-2})^2 \text{ m}^2 = 4\pi \times 10^{-4} \text{ m}^2$$

turns, $N_1 = 900$

turns, $N_2 = 45$

μ_r for iron = 800

$$\therefore M = \frac{\mu_0 \mu_r N_1 N_2 A}{l} = \frac{4\pi \times 10^{-7} \times 800 \times 900 \times 45 \times 4\pi \times 10^{-4}}{0.2\pi}$$

$$= 8.14 \times 10^{-2} \text{ Henry (Ans)}$$